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A multi-stage framework for cost-optimal predictive maintenance: From cost-sensitive classification to dynamic survival modeling: A comparative analysis using the Scania component X dataset

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Abstract

This study presents a comprehensive comparative analysis of maintenance strategies for heavy-duty trucks using the SCANIA Component X dataset. We implement and evaluate multiple machine learning approaches, including Logistic Regression, Multi-Layer Perceptron (MLP) Neural Networks, LightGBM, Random Forest, and survival analysis methods (Cox Proportional Hazards and Random Survival Forest). Our analysis focuses on cost-sensitive decision-making, recognizing the asymmetric costs of false negatives (missed failures) versus false positives (unnecessary maintenance). The results demonstrate that cost-sensitive approaches achieve up to 12.2% cost reduction compared to baseline strategies, with the Economic threshold-based strategy achieving an optimal balance between maintenance coverage and cost efficiency. The Random Forest model achieved 96.7% accuracy, while LightGBM achieved the lowest total cost at $52,364 among classification models. Survival analysis models showed strong discriminative ability, with Random Survival Forest achieving a concordance index of 0.895.

Keywords

predictive maintenance, machine learning, cost-sensitive learning, SCANIA, heavy-duty trucks, Random Forest, survival analysis

1. Introduction

Predictive maintenance (PdM) has emerged as a transformative approach in the automotive industry, leveraging advanced analytics and machine learning to anticipate component failures before they occur (Kharazian et al., 2024). Unlike

traditional reactive maintenance strategies that address failures after they happen, or preventive maintenance that follows fixed schedules regardless of actual component condition, predictive maintenance uses real-time operational data to optimize maintenance timing and resource allocation.

The economic implications of maintenance strategy selection are substantial. In the heavy-duty trucking industry, unexpected breakdowns can result in significant costs including emergency repairs, towing expenses, cargo delays, and potential safety hazards (Moat and Coleman, 2021). Conversely, excessive preventive maintenance leads to unnecessary part replacements and vehicle downtime. This creates a fundamental trade-off that cost-sensitive machine learning approaches aim to optimize.

The SCANIA Component X dataset represents a significant contribution to the predictive maintenance research community, providing real-world multivariate time series data from an internationally recognized truck manufacturer (Kharazian et al., 2024). This dataset addresses a critical gap in the field, as Original Equipment Manufacturers (OEMs) typically maintain strict confidentiality over operational data.

This study makes several contributions: First, we implement and compare multiple classification algorithms evaluating their performance using both traditional accuracy metrics and cost-based evaluation frameworks. Second, we develop and analyze cost-sensitive decision policies that explicitly account for the asymmetric costs of different error types. Third, we provide practical recommendations for maintenance strategy selection.

2. Related work

2.1 Predictive Maintenance in Heavy-Duty Vehicles

The application of machine learning to predictive maintenance has grown substantially in recent years. Ranasinghe et al. (2019) developed prognostic methodologies for scenarios with limited failure data availability. The SCANIA Air Pressure System (APS) dataset has served as an important benchmark (UCI Machine Learning Repository, 2017). Numerous studies have leveraged this dataset including work on handling missing values (Rafsunjani et al., 2019), cost-sensitive classification (Syed et al., 2020), and deep learning approaches (Taghandiki and DallakehNejad, 2023).

2.2 Cost-Sensitive Learning

Traditional classification metrics such as accuracy can be misleading in maintenance contexts where the costs of different error types vary substantially (Beikmohammadi et al., 2024). False negatives typically incur much higher costs than false positives. Akarte and Hemachandra (2018) applied boosting methods with cost-sensitive modifications. Beikmohammadi et al. (2024) developed a

cost-sensitive transformer model specifically designed for prognostics under highly imbalanced industrial data conditions.

2. Dataset description

3.1 SCANIA Component X Dataset Overview

The SCANIA Component X dataset comprises operational data collected from heavy-duty trucks, including multivariate time series readouts, repair records, and vehicle specifications (Kharazian et al., 2024). The training set contains 1,122,452 observations from 23,550 unique vehicles with 107 columns. The operational data includes 14 anonymized variables organized as numerical counters and multi-dimensional histograms.

3.2 Class Structure and Imbalance

The classification task involves predicting urgency level: Class 0 (>48 time steps before failure), Class 1 (24-48), Class 2 (12-24), Class 3 (6-12), and Class 4 (0-6, imminent failure). As shown in Figure 1, the dataset exhibits significant class imbalance with approximately 97% of observations belonging to Class 0 (No Issue). This imbalance reflects real-world conditions where most components operate normally at any given time, but poses significant challenges for machine learning models attempting to detect the minority failure classes.

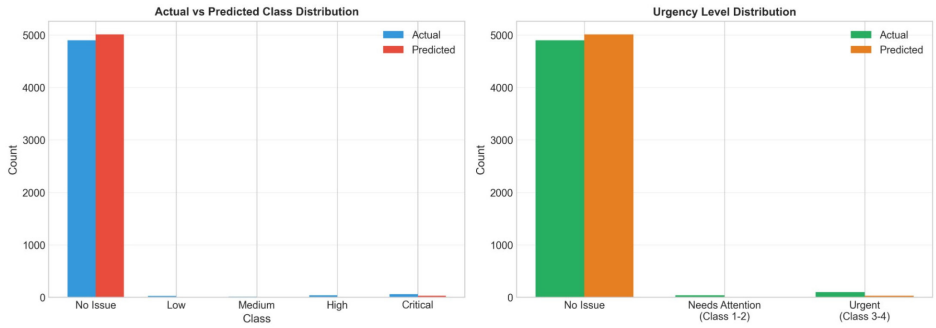


Figure 1. Actual vs Predicted Class Distribution showing severe class imbalance with ~97% Class 0 (No Issue). The right panel groups classes into urgency levels.

3.3 Cost Matrix

A key contribution of the SCANIA dataset is the provision of an expert-defined cost matrix that quantifies the economic impact of different prediction outcomes. The cost of missing an imminent failure (Class 4 predicted as Class 0) is \$500, representing potential breakdown costs, emergency repairs, and logistical disruptions. In contrast, a false positive (Class 0 predicted as Class 4) costs only \$10, representing the cost of an unnecessary workshop inspection. This 50:1 cost ratio fundamentally shapes optimal decision strategies.

Table 1. Cost matrix for prediction outcomes (USD)

	PRED 0	**PRED 1**	**PRED 2**	**PRED 3**	**PRED 4**
ACTUAL 0	0	7	8	9	10
ACTUAL 1	200	0	7	8	9
ACTUAL 2	300	200	0	7	8
ACTUAL 3	400	300	200	0	7
ACTUAL 4	500	400	300	200	0

4. Methology

4.1 Classification Models

We implemented four classification algorithms: (1) Logistic Regression - a linear classifier providing interpretable coefficients; (2) Multi-Layer Perceptron (MLP) Neural Network with two hidden layers and dropout regularization; (3) LightGBM - a gradient boosting framework using histogram-based algorithms; and (4) Random Forest - an ensemble method providing feature importance rankings.

4.2 Survival Analysis Models

We implemented Cox Proportional Hazards and Random Survival Forest (RSF) models. These methods model time-to-failure directly, enabling nuanced risk assessment while properly handling censored observations. Figure 2 presents the survival analysis results, including mean survival curves, risk score distributions, model performance comparison (C-index), and feature importance for RSF.

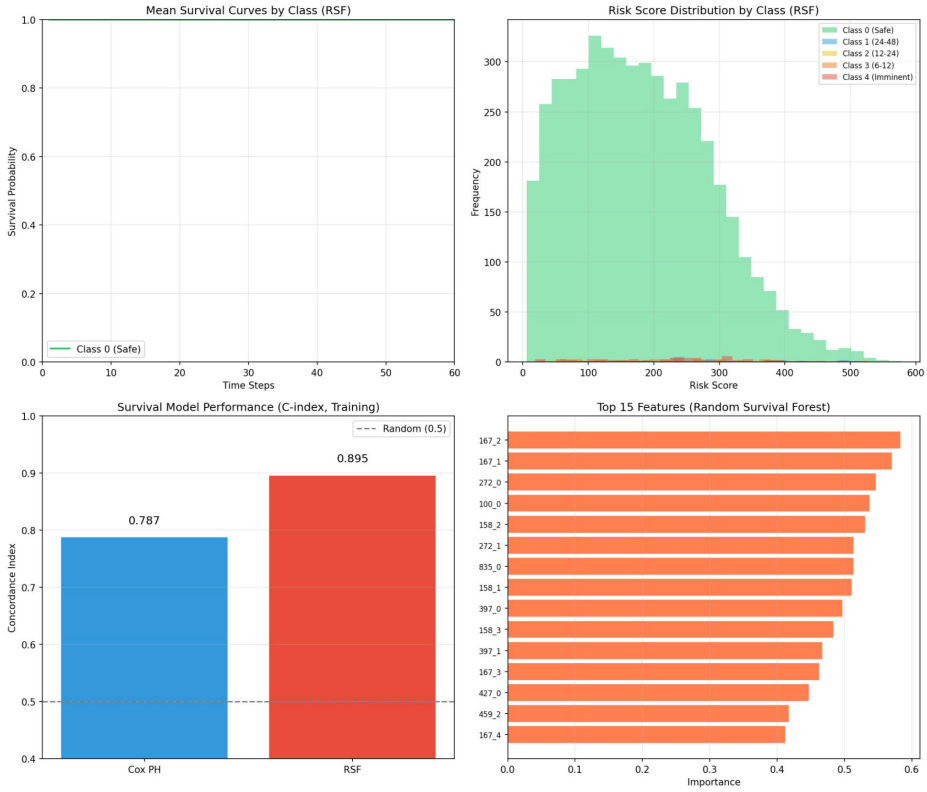


Figure 2. Survival analysis results: (a) Mean survival curves by class showing probability of component survival over time; (b) Risk score distribution by urgency class; (c) C-index comparison showing RSF (0.895) outperforming Cox PH (0.787); (d) Top 15 features for Random Survival Forest.

4.3 Maintenance Strategies

We evaluated four strategies: (1) Do Nothing - baseline with no predictive maintenance; (2) Model Prediction - direct use of predicted class labels; (3) Maintain All - conservative strategy scheduling maintenance for all vehicles; and (4) Cost-Sensitive (Economic) - threshold-based approach optimizing expected cost by considering prediction uncertainties and asymmetric costs.

5. Results

5.1 Model Performance Comparison

Figure 3 presents a comprehensive comparison of all models evaluated. A striking finding is the disconnect between accuracy and cost performance. As

shown in the accuracy vs cost trade-off plot (top right), MLP Neural Network and Random Forest achieve the highest accuracy (~97%), yet incur higher total costs than LightGBM (90.2% accuracy). This paradox occurs because high-accuracy models tend to predict the majority Class 0, missing critical failure cases that carry high misclassification costs.

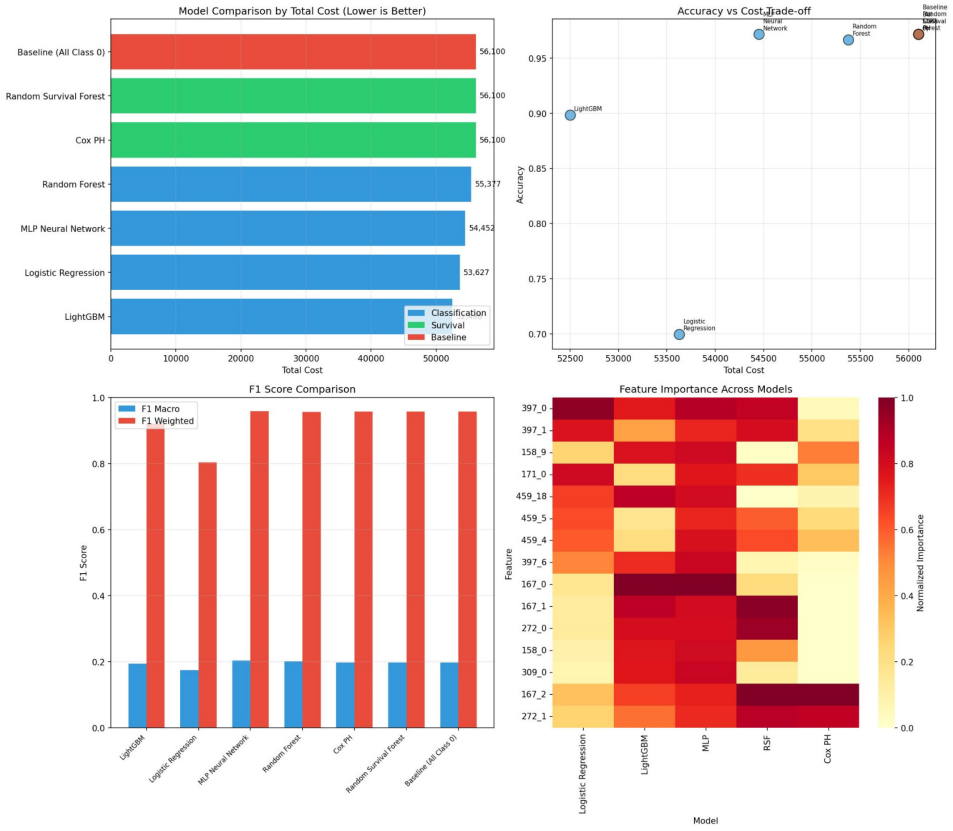


Figure 3. Comprehensive model comparison: (a) Total cost comparison showing LightGBM achieves lowest cost; (b) Accuracy vs Cost trade-off revealing the paradox where highest accuracy models incur higher costs; (c) F1 Score comparison; (d) Feature importance heatmap across models showing consistent patterns.

Table 2. Classification model performance comparison

MODEL	ACCURACY	TOTAL COST	F1 WEIGHTED
LOGISTIC REGRESSION	70.0%	$53,627	0.80
MLP NEURAL NETWORK	97.3%	$54,452	0.97
LIGHTGBM	90.2%	$52,364	0.94
RANDOM FOREST	96.7%	$55,377	0.97

5.2 Confusion Matrix and Cost Analysis

Figure 4 displays the confusion matrix alongside the associated cost distribution. The left panel shows that the model correctly classifies 4,876 Class 0 instances but misclassifies most failure cases (Classes 1-4) as Class 0. The right panel translates these misclassifications into costs, revealing that missed Class 4 failures (59 instances × $500 = $29,500) contribute the largest portion of total cost, followed by missed Class 3 failures ($16,400).

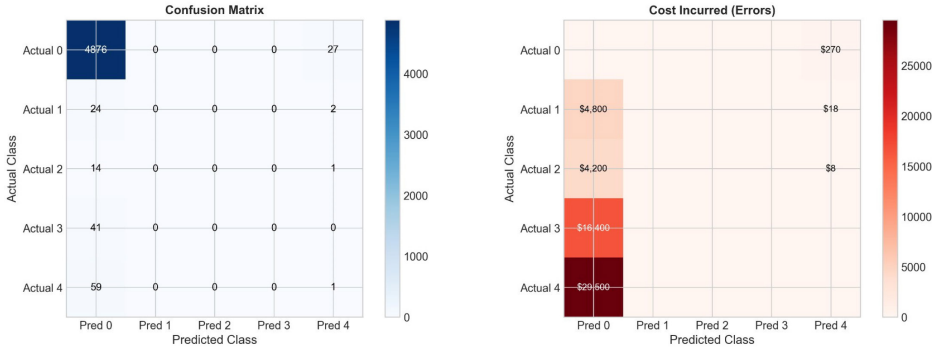


Figure 4. Confusion matrix (left) and associated cost incurred by errors (right). The cost heatmap shows that false negatives for Classes 3-4 (missed urgent failures) dominate total costs despite being numerically rare.

5.3 Maintenance Strategy Comparison

Figure 5 compares the four maintenance strategies on the test set of 5,045 vehicles. The Cost-Sensitive (Economic) strategy achieves the lowest total cost ($49,239), representing a 12.2% reduction compared to the Do Nothing baseline ($56,100). Notably, the simple Maintain All strategy achieves $49,671,

capturing 94% of the optimal savings with significantly less implementation complexity.

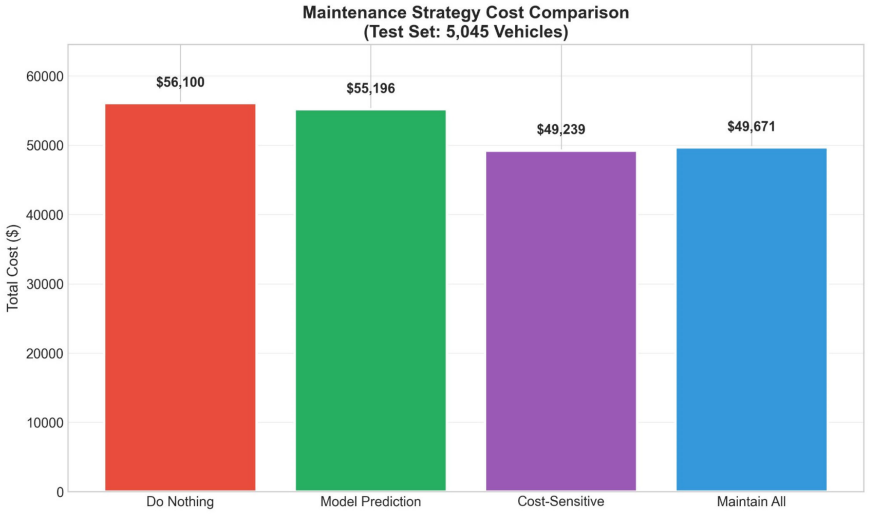


Figure 5. Maintenance strategy cost comparison on test set (5,045 vehicles). Cost-Sensitive strategy achieves optimal $49,239, while Maintain All achieves nearly equivalent $49,671.

Table 3. Maintenance strategy comparison

STRATEGY	TOTAL COST	SAVINGS VS BASELINE
DO NOTHING (BASELINE)	$56,100	-
MODEL PREDICTION	$55,196	1.6%
MAINTAIN ALL	$49,671	11.5%
COST-SENSITIVE (ECONOMIC)	$49,239	12.2%

5.4 Decision Policy Analysis

Figure 6 provides detailed analysis of the decision policy. The top-left panel shows that 5,014 vehicles require no action while only 31 are classified as critical. The bottom-left panel displays the distribution of net maintenance benefit, with the decision threshold (red dashed line) separating vehicles where maintenance provides positive expected value. The bottom-right confusion matrix confirms the model's strong performance on Class 0 but limited ability to detect minority failure classes.

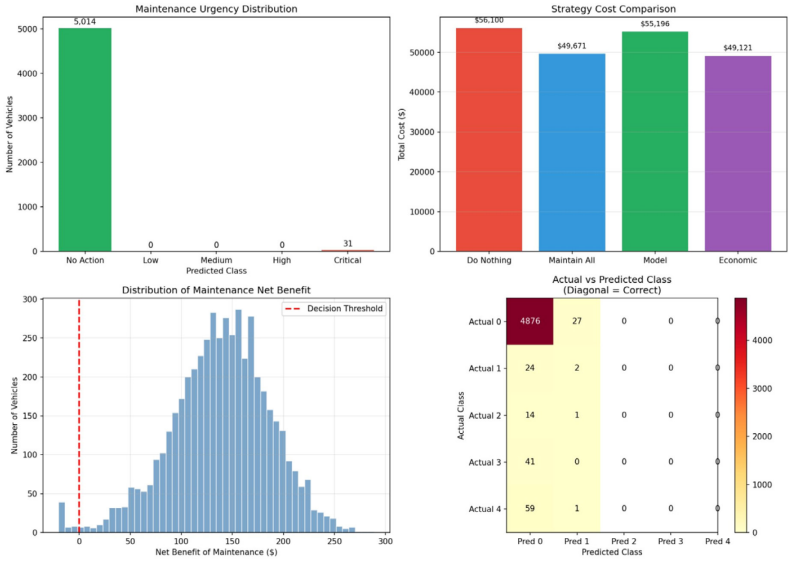


Figure 6. Decision policy analysis: (a) Maintenance urgency distribution; (b) Strategy cost comparison; (c) Net benefit distribution with decision threshold; (d) Detailed confusion matrix.

5.5 Feature Importance Analysis

Figure 7 presents the top 20 most important features from the Random Forest model. Features from histogram variable "167" (bins 1, 0, and 6) emerge as the most predictive, followed by multiple bins from variable "397". This pattern is consistent across different model architectures (as shown in Figure 3d), validating the robustness of these operational parameters as failure indicators.

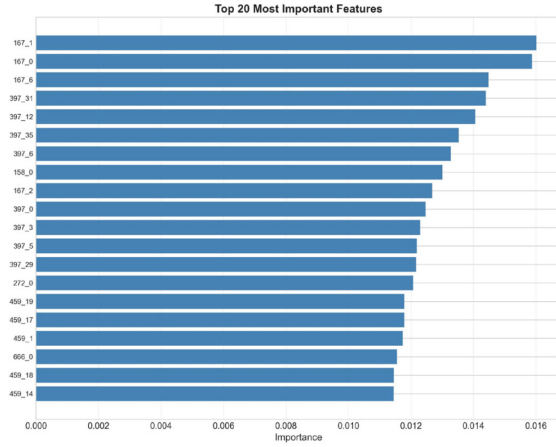


Figure 7. Top 20 most important features from Random Forest. Variable "167" bins dominate, followed by variable "397" bins, suggesting these operational parameters are key failure predictors.

6. Discussion

Our findings highlight several important insights for predictive maintenance practice:

First, the accuracy-cost disconnect demonstrates that standard machine learning metrics can be misleading in industrial applications where error costs are asymmetric. As shown in Figure 3, models optimized for accuracy (MLP, Random Forest at ~97%) perform worse on cost metrics than LightGBM at 90.2% accuracy. This occurs because accuracy-optimized models favor the majority class, missing critical failures.

Second, the strong performance of the simple Maintain All strategy (capturing 94% of optimal savings) suggests that in scenarios with highly asymmetric costs (50:1 ratio), conservative approaches may be near-optimal. The incremental benefit of sophisticated cost-sensitive optimization (0.7% additional savings) must be weighed against implementation complexity.

Third, feature importance analysis reveals consistent patterns across models, with histogram variables "167" and "397" emerging as dominant predictors. This convergence validates the robustness of the identified predictive signals and provides actionable insights for maintenance diagnostics.

Limitations include: the cost matrix based on expert estimates may vary across contexts; the dataset represents a specific subset of SCANIA trucks; and variable anonymization limits domain knowledge incorporation.

Conclusions

This study presented a comprehensive analysis of cost-sensitive machine learning approaches for predictive maintenance using the SCANIA Component X dataset. Key findings include:

(1) Traditional accuracy metrics can be misleading - LightGBM achieved lowest cost ($52,364) despite not having highest accuracy (90.2% vs 97.3% for MLP).

(2) Cost-sensitive strategies achieve up to 12.2% cost reduction compared to baseline, with the Economic strategy achieving optimal $49,239.

(3) Simple conservative strategies (Maintain All at $49,671) capture 94% of optimal savings, offering practical value without sophisticated optimization.

(4) Random Survival Forest achieves strong discriminative ability (C-index 0.895), substantially outperforming Cox Proportional Hazards (0.787).

Future work should integrate survival analysis predictions into cost-sensitive decision frameworks, explore online learning approaches to handle distribution drift, and investigate additional data sources such as telematics, environmental factors, and driver behavior patterns.

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